Machine Learning Interview Questions and Answers

1. What is Machine Learning, and how does it differ from traditional programming?

Machine Learning (ML) is a branch of AI that allows computers to learn patterns from data and make predictions without being explicitly programmed for every situation.

Interview answer:

Machine Learning is a branch of AI that enables computers to learn from data and make predictions. In traditional programming, we provide rules and data to get outputs, whereas in Machine Learning, algorithms learn patterns from data to make predictions.

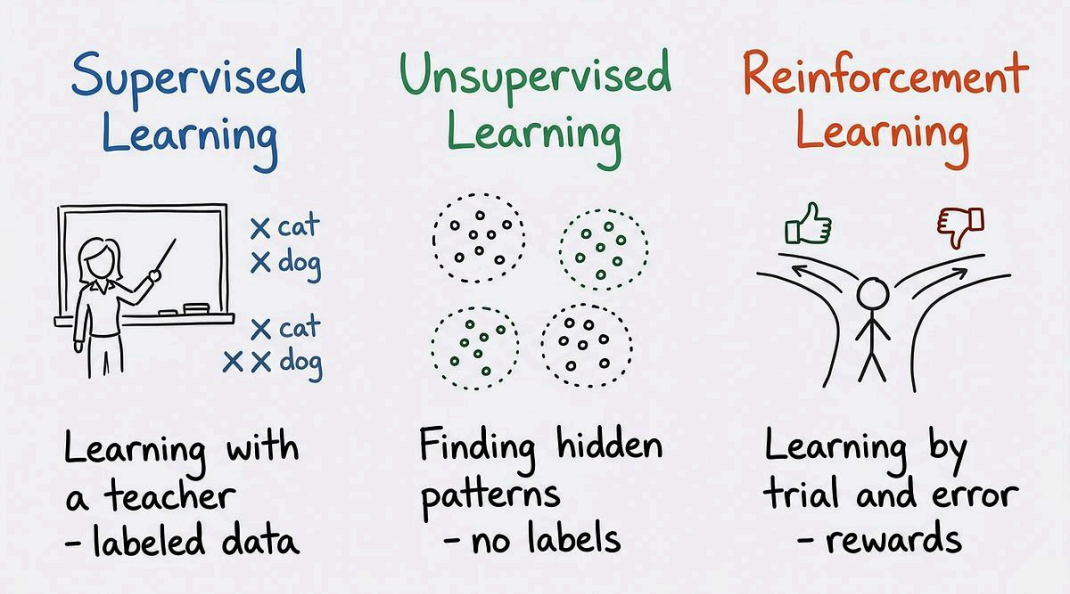
Example:

- Traditional programming: We manually write rules to identify spam emails.
- Machine Learning: We train a model using spam and non-spam emails.

2. What are the main types of Machine Learning?

There are three main types of Machine Learning:

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning



Interview answer:

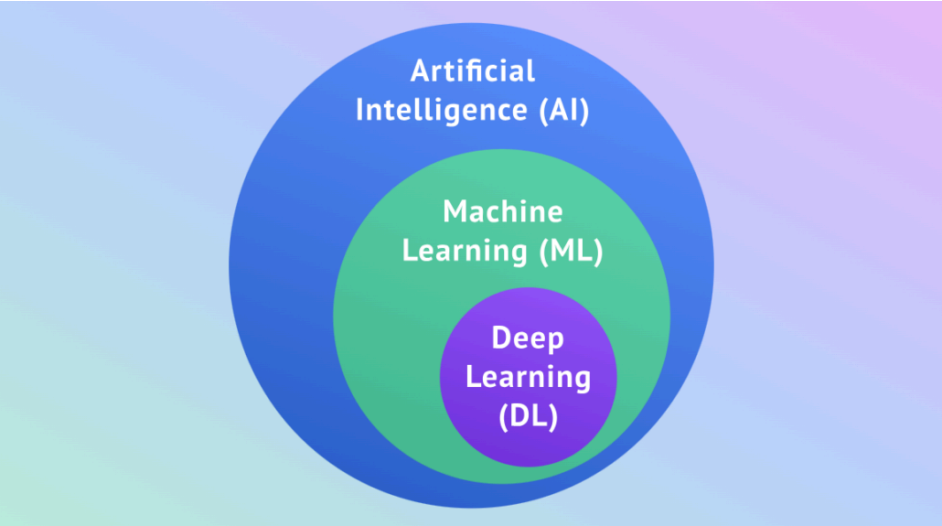
The three main types of Machine Learning are supervised learning, unsupervised learning and reinforcement learning. They differ based on how the model learns from data or interacts with its environment.

Example:

- Supervised: Predicting house prices.
- Unsupervised: Grouping customers based on purchasing behaviour.
- Reinforcement: Training a robot to navigate an environment.

3. What is the difference between Artificial Intelligence, Machine Learning and Deep Learning?

- Artificial Intelligence (AI): Making machines perform tasks that normally require human intelligence.
- Machine Learning (ML): A subset of AI where machines learn patterns from data.
- Deep Learning (DL): A subset of ML that uses neural networks with multiple layers.



Interview answer:

Artificial Intelligence is the broader concept of making machines intelligent. Machine Learning is a subset of AI that learns from data, and Deep Learning is a subset of Machine Learning that uses multi-layer neural networks.

Example:

- AI: A smart virtual assistant.
- ML: An email spam detection model.

- DL: A facial recognition system.

4. What is the difference between supervised, unsupervised and reinforcement learning?

Type	How it learns	Example
Supervised	Uses labelled data	House price prediction
Unsupervised	Uses unlabelled data	Customer segmentation
Reinforcement	Learns through rewards and penalties	Game-playing AI

Interview answer:

Supervised learning uses labelled data to make predictions. Unsupervised learning finds hidden patterns in unlabelled data. Reinforcement learning learns through interaction with an environment using rewards and penalties.

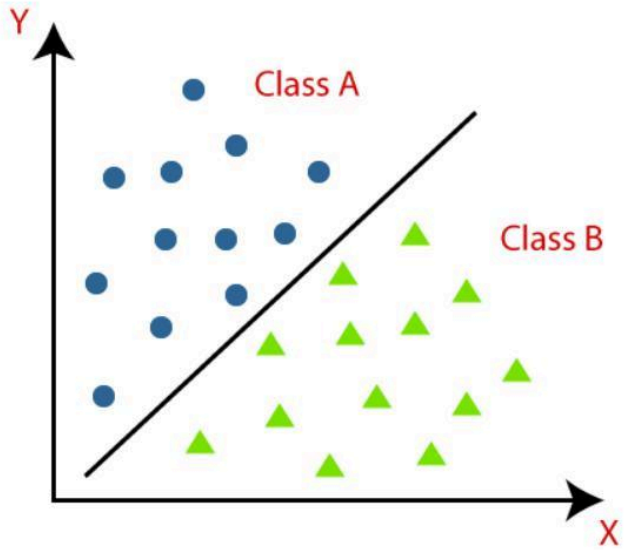
Easy way to remember:

- Supervised = Learn with answers.
- Unsupervised = Find patterns.
- Reinforcement = Learn through experience.

5. What is the difference between classification and regression?

Both are supervised Machine Learning techniques.

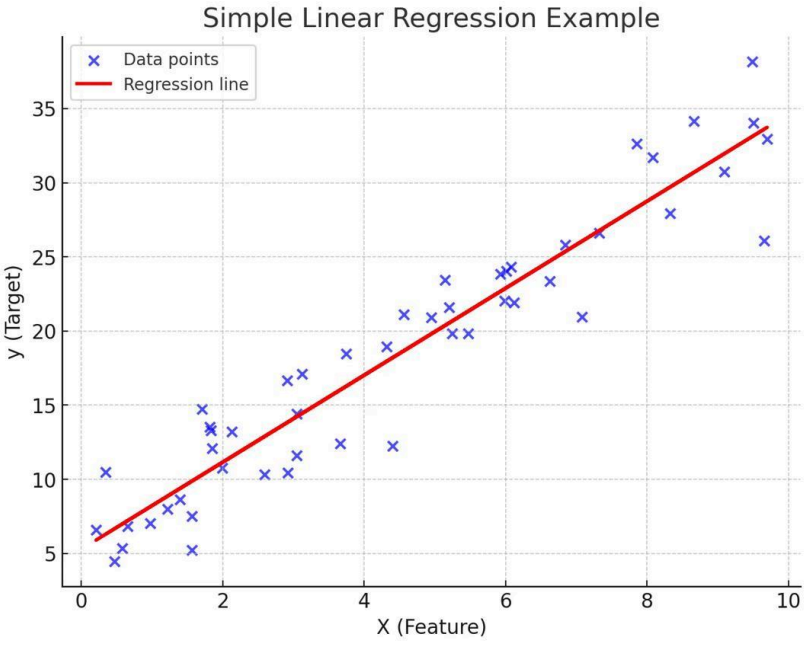
- Classification: Predicts a category or class.
- Regression: Predicts a continuous numerical value.



Classification

Predicts categories

Example: Spam / Not Spam



Regression

Predicts numbers

Example: House price ₹50 lakh

Interview answer:

Classification predicts discrete categories, whereas regression predicts continuous numerical values. Both are supervised learning techniques.

Examples:

- Classification: Predicting whether a student will pass or fail.
- Regression: Predicting a student's exam score.

6. What is the difference between training, validation and test datasets?

When building an ML model, we usually divide the dataset into three parts.

Dataset	Purpose
Training	Used to teach the model
Validation	Used to tune and improve the model
Testing	Used to evaluate final performance



Interview answer:

Training data is used to train the model, validation data is used to tune hyperparameters and select the best model, and test data is used to evaluate the final model on unseen data.

Example:

Suppose we have 1,000 records:

- Training: 700 records
- Validation: 150 records
- Testing: 150 records

These percentages are illustrative; the actual split depends on the project.

7. What are features and target variables in a Machine Learning dataset?

- Features (X): Input columns used by the model to make predictions.
- Target (y): The output column that the model tries to predict.

Example: Student Performance Dataset

Hours Studied	Attendance	Previous Marks	Final Result
5	90%	75	Pass
2	60%	40	Fail
7	95%	85	Pass

Interview answer:

Features are the input variables used by a Machine Learning model to make predictions, while the target variable is the output we want the model to predict.

Example: To predict house prices, house size, location and number of bedrooms are features, while house price is the target.

8. What is the difference between a model, an algorithm and a parameter?

These three terms are related but have different meanings.

Term	Meaning	Example
Algorithm	Learning method	Linear Regression
Model	Trained result	A trained house price predictor
Parameter	Value learned during training	Coefficients and intercept

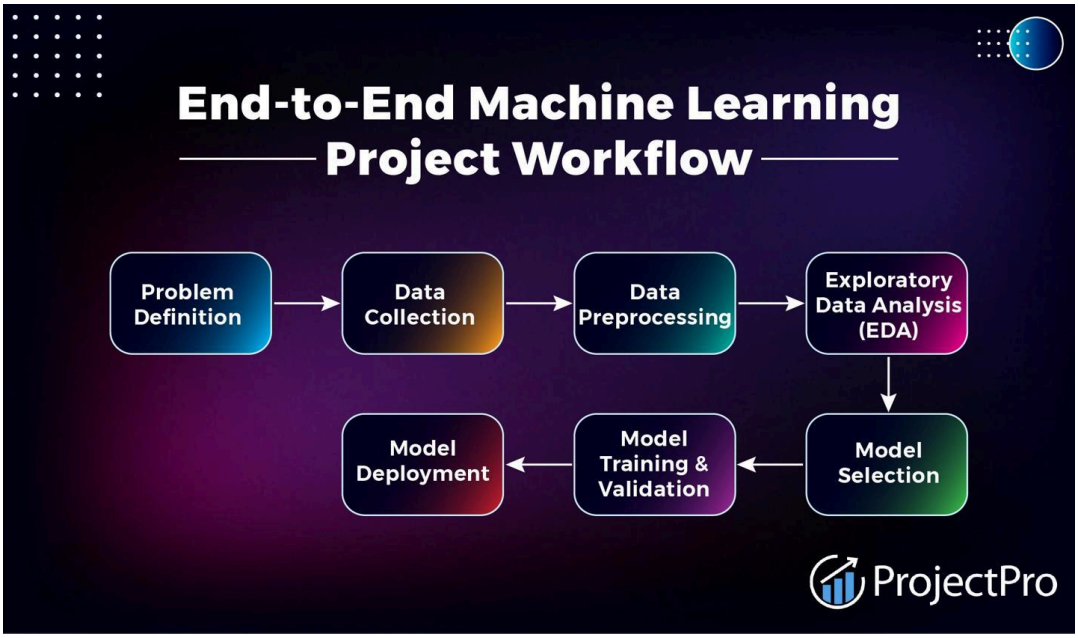
Interview answer:

An algorithm is the learning method used to train a model. A model is the trained system that makes predictions, and parameters are the values learned by the model during training.

Example: Linear Regression is an algorithm. After training, it learns coefficients and an intercept, which form part of the trained model.

9. What are the main stages of a typical Machine Learning project lifecycle?

A Machine Learning project follows several steps, from understanding the problem to deploying the model.



Main stages:

1. Problem Definition: Understand what needs to be predicted.
2. Data Collection: Collect relevant data.
3. Data Preprocessing: Handle missing values, duplicates and incorrect data.
4. Exploratory Data Analysis (EDA): Understand patterns in the data.
5. Feature Engineering: Create and select useful features.
6. Model Training: Train the algorithm using data.
7. Model Evaluation: Check model performance.
8. Deployment: Make the model available for real-world use.
9. Monitoring: Track performance and update the model when necessary.

Interview answer:

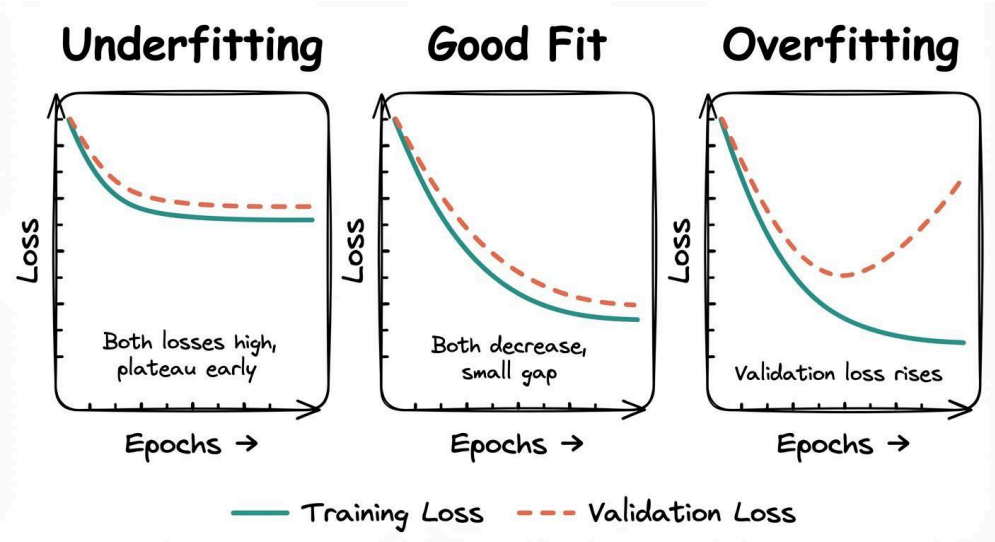
The main stages of a Machine Learning project are problem definition, data collection, data preprocessing, EDA, feature engineering, model training, evaluation, deployment and monitoring.

Example: For a house price prediction project, we collect house data, clean it, train a regression model, evaluate its accuracy and deploy it to predict prices for new houses.

10. What are the common challenges faced when building Machine Learning models?

Some common challenges include:

Challenge	Explanation
Missing Data	Some values are unavailable.
Overfitting	Model performs well on training data but poorly on unseen data.
Underfitting	Model fails to learn important patterns.
Imbalanced Data	Some classes have significantly more records than others.
Poor Data Quality	Data contains errors, duplicates or incorrect values.
Feature Selection	Choosing the most useful input variables.
Model Selection	Choosing a suitable algorithm.



Interview answer:

Common challenges in Machine Learning include missing data, poor data quality, overfitting, underfitting, imbalanced datasets, feature selection and choosing the right algorithm. These challenges can affect model performance and prediction accuracy.

Example: If a model memorises training data instead of learning general patterns, it may perform poorly on new data. This is called overfitting.

11. Why is data preprocessing important before training a Machine Learning model?

Data preprocessing is the process of cleaning and preparing raw data before giving it to a Machine Learning model.

Raw data may contain:

- Missing values
- Duplicate records
- Incorrect data
- Outliers
- Different data formats and scales

Interview answer:

Data preprocessing is important because raw data may contain missing values, duplicates, outliers and inconsistent formats. It helps improve data quality and prepares the dataset for better Machine Learning performance.

Example: Before predicting house prices, we handle missing house sizes, remove duplicate records and convert location names into numerical values.

12. How do you identify and handle missing values in a dataset?

Missing values are data entries that are empty or unavailable.

We can identify them using Pandas:

```
None
import pandas as pd

df = pd.read_csv("data.csv")
```

```
# Count missing values in each column
print(df.isnull().sum())

# Check total missing values
print(df.isnull().sum().sum())
```

Common ways to handle missing values:

Method	When to use
Remove rows	Few rows have missing values and removal is appropriate
Mean	Numerical data without strong skew
Median	Numerical data with skew or outliers
Mode	Categorical data
KNN Imputation	Estimate missing values using similar records

Interview answer:

We can identify missing values using `isnull()` or `isna()` in Pandas. We can handle them by removing rows or columns, or filling them using mean, median, mode or imputation techniques such as KNNImputer.

Example:

Original ages:

```
20, 25, NaN, 30, 35
```

Using mean imputation:

$$Mean = \frac{20+25+30+35}{4} = 27.5$$

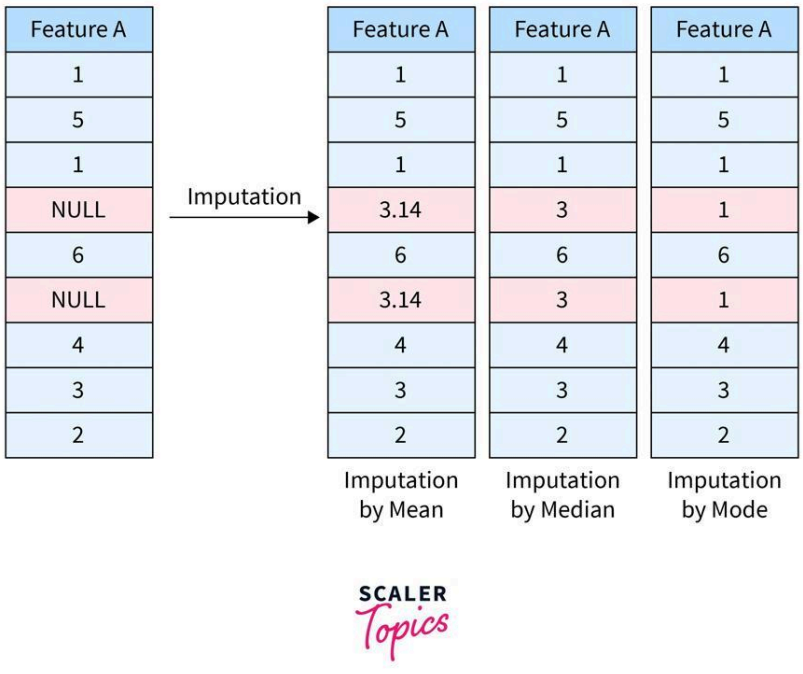
New data:

```
20, 25, 27.5, 30, 35
```

13. What is the difference between SimpleImputer and KNNImputer in Scikit-learn?

Both are used to handle missing values, but they work differently.

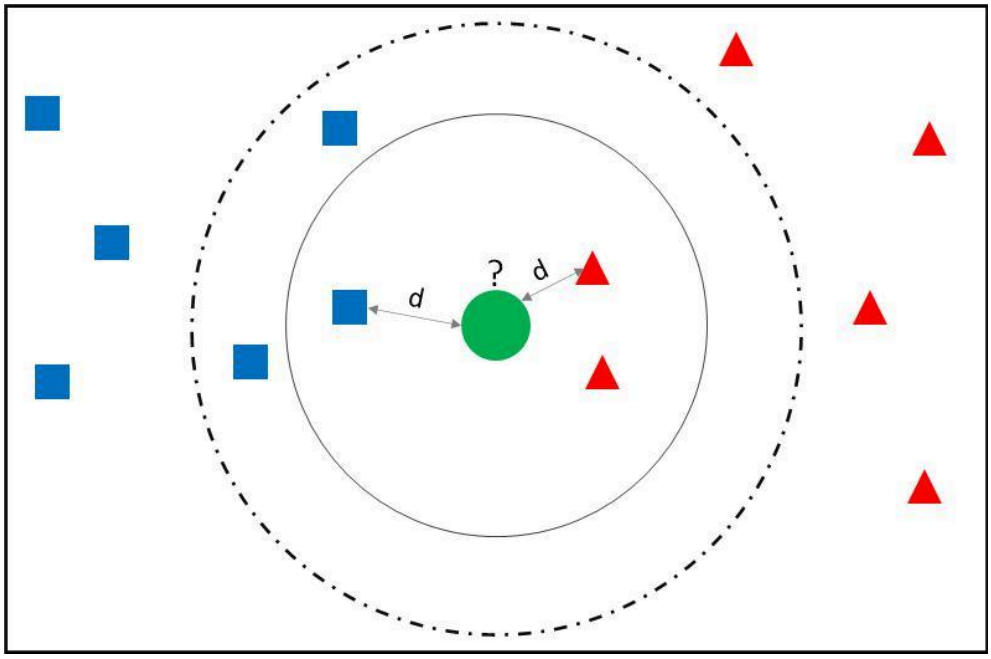
- SimpleImputer: Fills missing values using a simple statistical strategy such as mean, median or most frequent value.
- KNNImputer: Finds similar records (nearest neighbours) and estimates missing values using those records.



SCALER Topics

SimpleImputer

Uses column statistics.



KNNImputer

Uses similar records.

Code example:

SimpleImputer:

```
None
from sklearn.impute import SimpleImputer

imputer = SimpleImputer(strategy="mean")

df[["Age"]] = imputer.fit_transform(df[["Age"]])
```

KNNImputer:

```
None
from sklearn.impute import KNNImputer

imputer = KNNImputer(n_neighbors=3)

df[["Age", "Salary"]] = imputer.fit_transform(
```

```
    df[["Age", "Salary"]]
)
```

Interview answer:

SimpleImputer fills missing values using statistical methods such as mean, median or mode. KNNImputer fills missing values by considering similar data points using the K-nearest neighbours algorithm.

Key difference: SimpleImputer uses a column-level strategy, whereas KNNImputer considers relationships between records.

14. What is the difference between label encoding and one-hot encoding?

Both techniques convert categorical (text) data into numerical data.

Label Encoding: Assigns a unique integer to each category.

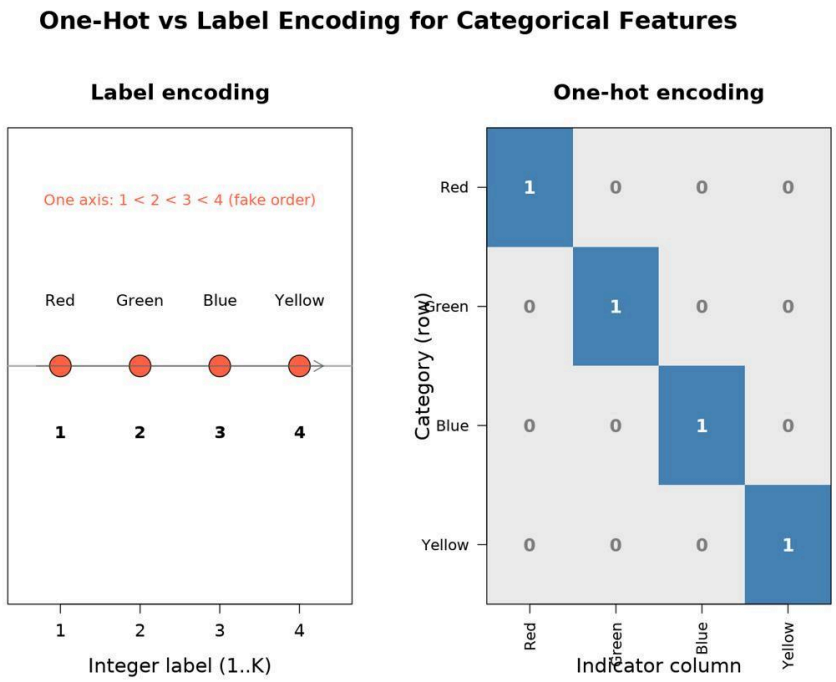
Example:

Colour	Encoded value
Red	0
Blue	1
Green	2

One-Hot Encoding: Creates separate columns for each category and assigns 0 or 1.

Colour	Red	Blue	Green

Red	1	0	0
Blue	0	1	0
Green	0	0	1



Code example:

Label Encoding:

```
None
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df["Gender"] = le.fit_transform(df["Gender"])
```

One-Hot Encoding:

```
None
from sklearn.preprocessing import OneHotEncoder

encoder = OneHotEncoder(
    sparse_output=False
)

encoded = encoder.fit_transform(
    df[["City"]]
)
```

Interview answer:

Label encoding converts categories into integer values, whereas one-hot encoding creates separate binary columns for each category. One-hot encoding is useful when categories do not have a natural order.

Example: For colours, one-hot encoding is generally more appropriate. For education levels such as School, Bachelor's and Master's, ordinal encoding may be more suitable because they have a natural order.

15. What is feature scaling, and why is it important?

Feature scaling is the process of bringing numerical features into a similar range.

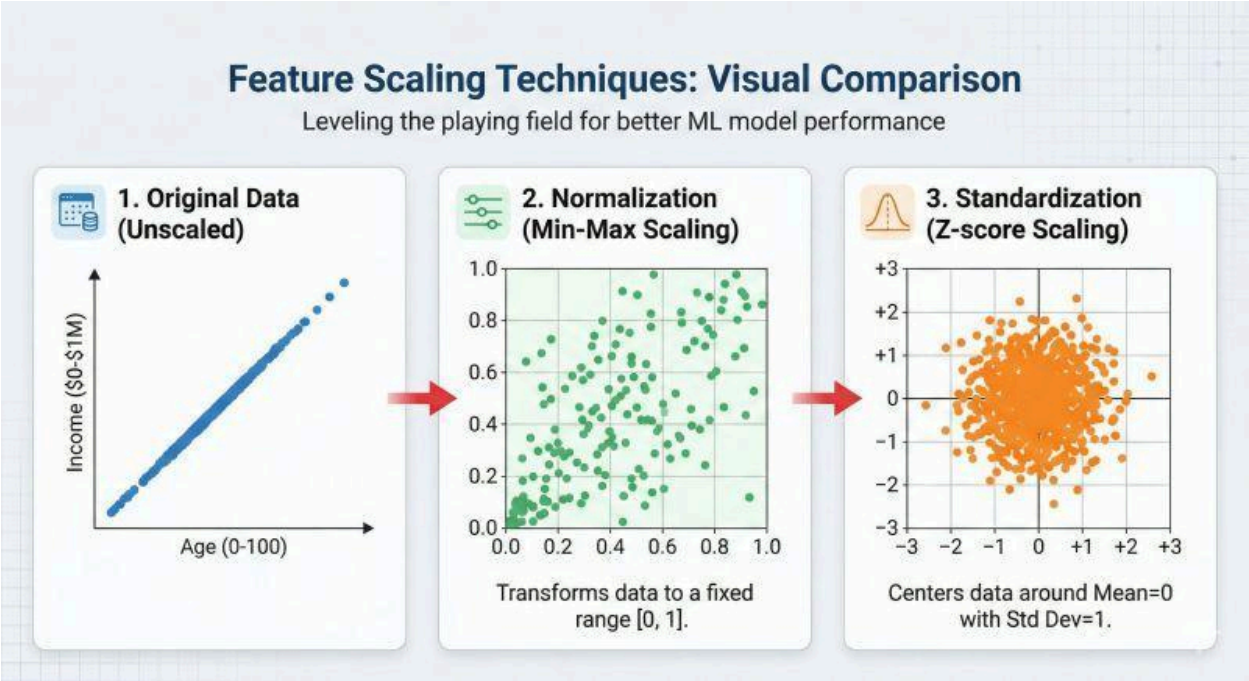
Consider this dataset:

Age	Salary
25	25,000
30	40,000
35	60,000

Age values are small, but salary values are much larger.

Some ML algorithms are sensitive to these differences in scale.

Visual understanding:



Common scaling methods:

1. Standardisation
2. Normalisation

Interview answer:

Feature scaling is the process of bringing numerical features into a similar range. It is important because algorithms such as KNN, K-Means and SVM are sensitive to differences in feature scales.

Example: Scaling age and salary helps prevent salary's larger numerical values from dominating distance calculations in KNN.

16. What is the difference between normalisation and standardisation?

Both are feature scaling techniques, but they use different formulas.

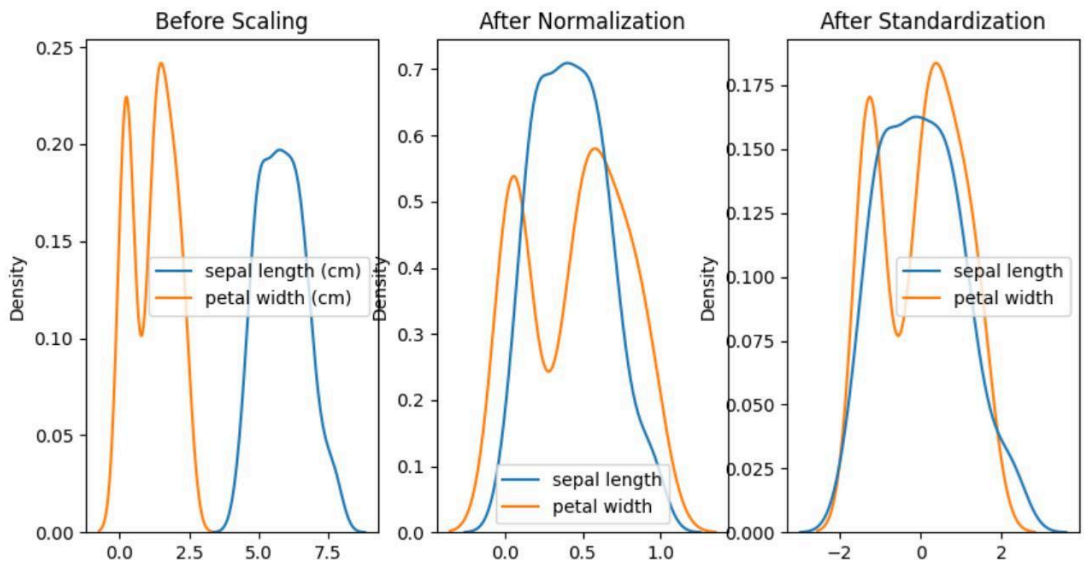
1. Normalisation (Min-Max Scaling)

Converts values into a specified range, commonly 0 to 1.

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

2. Standardisation (Standard Scaling)

Transforms values so that the resulting feature has a mean of approximately 0 and a standard deviation of approximately 1.



Code example:

Normalisation:

```
None
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
df[["Age", "Salary"]] = scaler.fit_transform(
    df[["Age", "Salary"]]
)
```

Standardisation:

```
None
from sklearn.preprocessing import StandardScaler
```

```
scaler = StandardScaler()
df[["Age", "Salary"]] = scaler.fit_transform(
    df[["Age", "Salary"]]
)
```

Interview answer:

Normalisation scales values into a fixed range, usually 0 to 1, whereas standardisation transforms values using the mean and standard deviation, resulting in a mean of 0 and standard deviation of 1.

Key difference:

Normalisation	Standardisation
Usually range 0–1	Mean 0, SD 1
Uses minimum and maximum	Uses mean and standard deviation
MinMaxScaler	StandardScaler

17. What is an outlier, and how can you detect and handle outliers?

An outlier is a data point that is significantly different from most other values in a dataset.

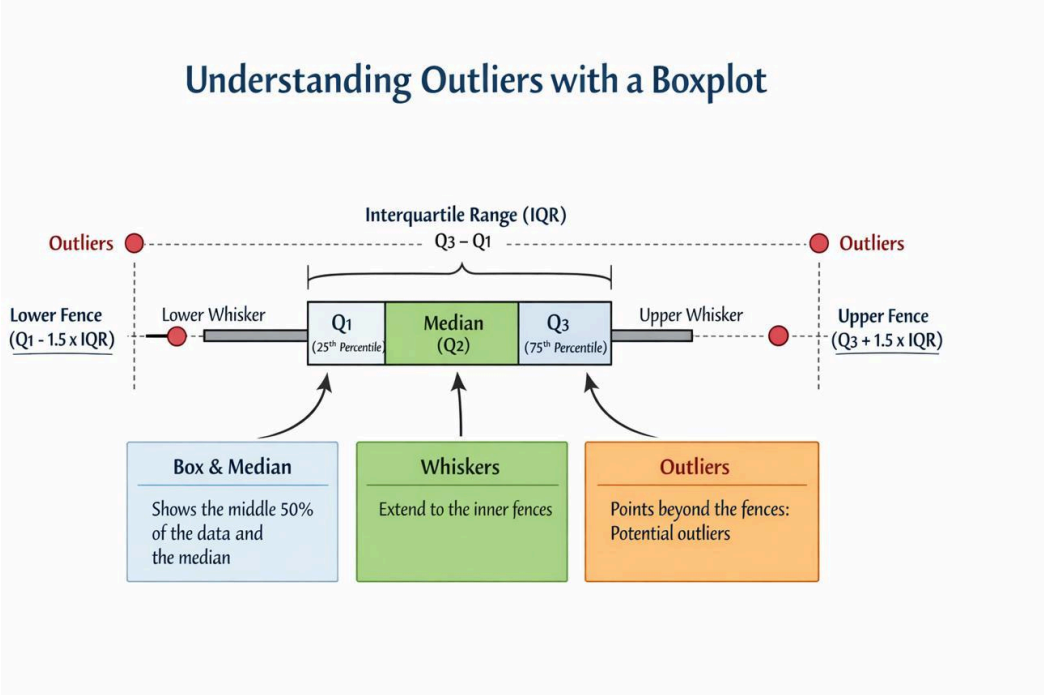
Example:

```
20, 22, 25, 27, 30, 150
```

Here, 150 may be an outlier because it is much higher than the other values.

Common detection methods:

1. Box Plot
2. IQR Method
3. Z-Score



IQR Method:

$$IQR = Q3 - Q1$$

Lower limit:

$$Q1 - 1.5 \times IQR$$

Upper limit:

$$Q3 + 1.5 \times IQR$$

Values outside these limits are potential outliers.

Code example:

```
None
Q1 = df["Salary"].quantile(0.25)
```

```
Q3 = df["Salary"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[
    (df["Salary"] < lower) |
    (df["Salary"] > upper)
]

print(outliers)
```

Ways to handle outliers:

- Remove incorrect or irrelevant records.
- Cap extreme values using Winsorisation.
- Apply transformations such as log transformation.
- Keep genuine outliers if they contain important information.

Interview answer:

An outlier is a value that is significantly different from other observations. We can detect outliers using box plots, IQR and Z-score methods. We can handle them by removing, capping or transforming extreme values, depending on the situation.

18. What is feature engineering? Give practical examples.

Feature engineering is the process of creating new features or transforming existing features to help a Machine Learning model learn better patterns.

Practical examples:

Existing data	New feature

Date of birth	Age
Purchase date	Month, Day, Year
Height and Weight	BMI
Total sales and quantity	Average selling price
First name and last name	Full name

Code example:

```
None
# Create age feature
df["Age"] = 2026 - df["Birth_Year"]

# Create BMI feature
df["BMI"] = (
    df["Weight"] / (df["Height"] ** 2)
)

# Extract month from date
df["Purchase_Date"] = pd.to_datetime(
    df["Purchase_Date"]
)

df["Month"] = df["Purchase_Date"].dt.month
```

Interview answer:

Feature engineering is the process of creating new features or transforming existing features to improve Machine Learning performance. Examples include calculating age from date of birth, creating BMI from height and weight, and extracting month and year from dates.

19. What is feature selection, and how does it differ from feature extraction?

Both techniques help reduce the number of features in a dataset.

Feature Selection:

Selects the most useful existing features and removes unnecessary features.

Example:

Original features:

```
Age, Salary, Height, Weight, Customer_ID
```

Selected features:

```
Age, Salary, Weight
```

The selected features remain unchanged.

Feature Extraction:

Creates new features by transforming or combining existing features.

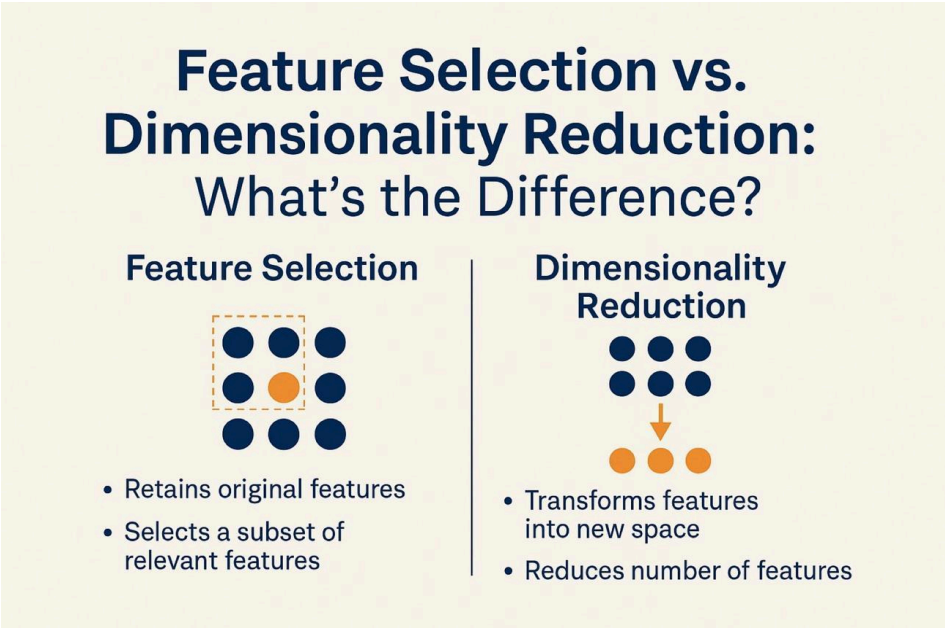
Example:

Original features:

```
Height, Weight
```

New extracted feature:

```
BMI
```



Interview answer:

Feature selection chooses the most important existing features from a dataset, whereas feature extraction transforms existing features into a new set of features.

Key difference:

Feature Selection	Feature Extraction
Selects existing features	Creates transformed features
Original features remain	New representations are created
Example: SelectKBest	Example: PCA

20. What is data leakage, and how can you prevent it during preprocessing?

Data leakage happens when information that should not be available during model training accidentally reaches the model.

This can make the model show excellent training or validation performance but perform poorly on genuinely unseen data.

Example:

Suppose we are predicting whether a student will pass an exam.

Features:

- Study hours
- Attendance
- Previous marks
- Final exam result

If we accidentally include the final exam result as an input feature, the model already knows the answer it needs to predict.

This is data leakage.

Incorrect approach

Training data + Test information
Data Leakage

Correct approach

Split dataset first
Fit preprocessing on training data only
Transform validation and test data

How to prevent data leakage:

1. Split the dataset before preprocessing.
2. Fit scalers and imputers only on training data.
3. Apply the same fitted transformations to validation and test data.
4. Remove features that reveal the target.
5. Use Scikit-learn Pipelines to organise preprocessing and model training.

Code example:

```
None
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler

X = df.drop("Target", axis=1)
y = df["Target"]

# Split data first
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Fit only on training data
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
# Only transform test data
X_test = scaler.transform(X_test)
```

Interview answer:

Data leakage occurs when information from outside the training data, especially validation or test data, is used during model training. It can lead to unrealistically high performance. We can prevent it by splitting the data first, fitting preprocessing techniques only on training data and using pipelines.